

NLP 2025-2026 Homeworkp

Gaia Citro Lucia Fornetti
Sapienza University of Rome

{citro.2026094, fornetti.2214370}@studenti.uniroma1.it

Abstract

Cross-lingual phonetic transliteration maps the orthography of a source language into an intuitive phonetic representation compliant with the target language’s phonotactics. In this work, we address the task of directly converting English words into Italian-readable phonetic graphemes using sequence-to-sequence architectures. We present a comparative benchmark between a Bidirectional LSTM with Bahdanau attention and a compact BART Transformer. Moreover, we evaluate the impact of source-side subword tokenization by contrasting Byte Pair Encoding (BPE) with a probabilistic Unigram model against a character-level decoder. Models are trained on a leakage-filtered dataset derived from the CMU Pronouncing Dictionary and assessed via Character Error Rate (CER). Experimental results show that BLSTM achieves the best performance with Unigram Tokenization, closely followed by the BLSTM architecture, using BPE. Our findings confirm the effectiveness of subword modeling in capturing irregular English grapheme-to-sound mappings.

1 Task Description / Problem Statement

Cross-lingual phonetic transliteration is the process of converting the written orthography of a source language into an intuitive phonetic representation using the graphemic and phonotactic rules of a target language. While conventional Grapheme-to-Phoneme (G2P) systems map words to formal, language-agnostic phonetic alphabets such as the International Phonetic Alphabet (IPA) or ARPABET, this task aims at producing a phonetic transcription directly readable by

native speakers of the target language. Specifically, given an input sequence of characters representing an English word $X = (x_1, x_2, \dots, x_N)$, the goal is to generate a sequence of Italian graphemes $Y = (y_1, y_2, \dots, y_M)$ that phonetically reproduces the correct English pronunciation when read according to standard Italian orthographic conventions.

1.1 Examples

For this task, once an English word is processed, it is transliterated into readable Italian graphemes approximating the pronunciation specified by the standard ARPABET system. Table 1 illustrates representative examples showcasing typical phonetic transformations, including silent letter reduction, diphthong reconstruction, and consonant adaptation.

Table 1: Representative transliteration examples showing the mapping from English source orthography to Italian phonetic output.

English (X)	ARPABET (Intermediate)	Target (Y)
<i>somewhere</i>	S AH1 M W EH2 R	samuer
<i>time</i>	T AY1 M	taim
<i>choir</i>	K W AY ER	cuair
<i>thought</i>	TH AO T	tot
<i>friendly</i>	F R EH1 N D L IY0	frendli

1.2 Real-world Applications

Downstream applications for cross-lingual phonetic transliteration span several key domain areas:

Cross-Lingual Text-to-Speech (TTS) on Monolingual Devices: Enables Italian-configured edge devices and smart assistants to seamlessly vocalize English content (e.g., mixed-language notifications, navigation) without the computational overhead of loading dedicated English TTS engines or polyglot acoustic models.

Computer-Assisted Language Learning (CALL) for Italian Learners: Acts as a pronunciation scaffold for Italian students. By converting opaque English orthography into transparent Italian phonetics, it helps learners overcome phonotactic hurdles (e.g., silent letters, vowel reduction) and practice reading aloud without needing to master standard IPA symbols.

Speech Pronunciation Aids for Public Speaking: Generates transparent phonetic scripts to assist Italian professionals (e.g., actors, researchers, diplomats) in delivering English speeches, presentations, and scripted remarks with accurate, natural-sounding pronunciation, even with limited language proficiency.

2 Related work

This section reviews key research shaping our study, focusing on architectural efficiency, the recurrent versus Transformer debate, and strategies for handling irregular vocabulary.

2.1 Architectural Paradigms and Efficiency

One trend in this research is the shift toward lightweight, autoregressive models [3]. A recent comparative study by Arabov [1] demonstrates that compact Transformers trained from scratch on limited datasets optimize computational costs and can outperform massive pre-trained models. Furthermore, Chary and Ramirez [3] highlight that purely autoregressive generation maintains phonetic precision without relying on larger teacher networks.

2.2 The Recurrent vs. Transformer Debate

There are two conflicting camps on the issue of optimal neural architectures. While Transformers can converge faster and manage linguistic asymmetries better than recurrent networks [1], the investigation by Route et al. [6] establishes that Bidirectional LSTMs (BLSTMs) with global attention remain a highly robust baseline. The state-of-the-art character-level accuracy achieved by this specific architecture [5] directly motivated our selection of the BLSTM-Att as the foundational baseline for this comparative study. Our findings align with this precedent: our BLSTM with attention outperformed the custom BART architecture during inference, proving its remarkable resilience on unseen data.

2.3 Task Formulation and Rare Vocabulary

There is overall debate regarding "end-to-end" joint models that predict phonemes, syllables, and stress simultaneously. There is overall debate regarding whether predicting everything at once really helps. Since doing so actually degrades performance for tricky languages like English [7], we chose to focus exclusively on phoneme prediction. Finally, Le et al. [4] demonstrate that purely graphemic networks fail significantly on low-frequency words and proper nouns. This firmly motivates our manually curated test set, designed with extreme phonetic irregularities to rigorously expose model weaknesses

3 Datasets and Benchmarks

Due to the absence of dedicated datasets for English-to-Italian phonetic transliteration, it was necessary to construct a custom synthetic benchmark to train and evaluate the models. The dataset generation pipeline builds upon the CMU Pronouncing Dictionary [2] as the primary source of phonetic transcriptions. For each English lexical entry, the corresponding sequence of ARPABET phonemes is extracted and converted into an Italian phonetic transliteration through a deterministic, rule-based mapping. The resulting pipeline outputs a structured JSONL dataset containing pairs of English source words and their corresponding Italian phonetic targets, which are subsequently used to train the neural architectures.

3.1 Deterministic Phoneme-to-Grapheme Mapping

The conversion algorithm processes the phonetic sequences by first stripping all numeric stress markers from the CMU representation. Subsequently, it applies a context-aware mapping strategy to translate English phonemes into Italian orthographic rules accurately.

While most vowels and unambiguous consonants follow a direct static mapping (as illustrated in Table 2), context-dependent logic, specifically a lookahead operation, was implemented to resolve the orthographic ambiguity of velar and affricate sounds:

- **Soft Affricates (*CH*, *JH*):** These phonemes are mapped to *c* and *g* when preceding front vowels (*EH*, *EY*, *IH*, *IY*, *ER*) or when located at the absolute end of a word (e.g., *fridge* →

Table 2: Subset of the custom ARPABET-to-Italian deterministic mapping.

ARPABET Symbol	Italian Grapheme	Example
G	gh	<i>green</i>
CH	ci	<i>cheese</i>
SH	sci	<i>she</i>
OW	ou	<i>oat</i>
AY	ai	<i>hide</i>

frig). When preceding back vowels or consonants, they are mapped to *ci* and *gi*.

- **Hard Velars (*K*, *G*):** These are mapped to *ch* and *gh* when preceding front vowels to maintain the hard sound in Italian orthography. Conversely, they are mapped to *c* and *g* when appearing at the end of a word, or before back vowels and consonants.

3.2 Dataset Generation and Preprocessing

The dataset creation pipeline parsed the raw dictionary by filtering out comments, annotations, and variant numbering (e.g., *word(2)*) to extract clean base orthographies. To prevent data leakage, test set words and morphologically related neighbors were manually excluded by adding a comment tags in the raw dictionary file. Next, an automated script applied deterministic mapping rules to generate a structured JSONL dataset pairing English words with their Italian phonetic transliterations and stress-free phonetic representations. The compiled data was split into an 80% training and 20% validation set, reserving the test set for a separate evaluation protocol. Finally, clean text corpora were exported to train SentencePiece tokenizers, processing the English inputs via Byte Pair Encoding (BPE) and Unigram strategies prior to neural network training.

4 Existing tools, libraries, papers with code

All code, datasets, and preprocessing pipelines developed for this project are publicly available on GitHub at [gaiacitro/Cross-Lingual-Phonetic-Transliteration](https://github.com/gaiacitro/Cross-Lingual-Phonetic-Transliteration). To provide access to sufficient computational resources and enable accelerated GPU training, all scripts and workflows have been orchestrated within an interactive Google Colab notebook, directly executable at colab.research.google.com/drive/1U2hLqS_U1mc61Toaph2MWMwwZC9Qkb-I.

4.1 Libraries and Software Frameworks

The models and workflows were developed using the following primary open-source libraries:

- **PyTorch:** Employed for implementing the BLSTM architecture with Bahdanau Attention, managing tensor operations, and defining custom training loops.
- **Hugging Face Transformers:** Utilized for configuring, training, and running inference on the BART sequence-to-sequence model.
- **SentencePiece:** Used to train and deploy subword tokenization models (BPE, Unigram) for English and the character-level model for Italian.
- **JiWER:** Employed for metric tracking and evaluation, specifically calculating the Character Error Rate (CER) across all experimental configurations.

5 State-of-the-Art Evaluation

Because cross-lingual orthographic-phonetic transliteration from English to transparent Italian graphemes is a novel task without an existing public benchmark, direct numerical baselines from prior literature are not available. However, our evaluation protocol is grounded in the standard assessment methodologies established across adjacent computational linguistics domains, specifically Grapheme-to-Phoneme (G2P) conversion, machine transliteration, and sequence-to-sequence phonetic modeling.

5.1 Alignment with Established Evaluation Metrics Protocols

Our chosen evaluation protocol aligns directly with state-of-the-art literature in Grapheme-to-Phoneme (G2P) conversion and cross-lingual grapheme mapping, where systems are traditionally assessed via string distance metrics computed at the sub-word and character level. Rather than relying on exact word-level accuracy (or Word Error Rate, WER), existing benchmarks predominantly adopt the Character Error Rate (CER) or Phoneme Error Rate (PER).

Exact word-level match criteria impose a strict binary penalty where any minor phonetic divergence (such as predicting an acoustically close alternate vowel grapheme) marks the entire sequence as incorrect. In contrast, character-level

Levenshtein distance metrics provide a continuous, fine-grained measure of edit operations (substitutions, deletions, and insertions), reflecting the phonetic plausibility and acoustic proximity of the predicted strings more faithfully.

The Character Error Rate is formally defined as:

$$CER = \frac{S + D + I}{N} \quad (1)$$

where S , D , and I denote the total counts of character-level substitutions, deletions, and insertions, respectively, and N represents the total length (number of characters) of the ground-truth reference sequence.

By applying this standardized metric across deterministic phonetic mappings derived from the CMU Pronouncing Dictionary, our experimental setup establishes a rigorous and reproducible baseline for cross-lingual Italian phonetic transliteration.

6 Comparative evaluation

To rigorously assess the impact of subword tokenization and network architecture on cross-lingual phonetic transliteration, we designed a comprehensive comparative experimental setting. This section outlines the data configurations used to test our tokenization hypotheses, describes the selected sequence-to-sequence systems alongside their baselines, and defines the metrics and protocol utilized to evaluate generation accuracy.

6.1 Dataset Configurations

To systematically compare source-side subword algorithms, the CMU-derived corpus is instantiated into two distinct dataset variations. Both variations pair subword-segmented English inputs with identical character-tokenized Italian targets, differing exclusively in their source tokenization strategy, which is applied strictly on a word-by-word basis:

- **Byte-Pair Encoding (BPE):** A deterministic, bottom-up approach that iteratively merges the most frequent adjacent character pairs, producing static, frequency-driven subword fragments.
- **Unigram Language Model:** A probabilistic, top-down approach that starts with a large base vocabulary and iteratively trims

it. It allows for subword regularization (sampling multiple segmentations for the same word), providing a dynamic and phonotactically flexible representation.

Applying tokenization word-by-word preserves Unigram’s dynamic probabilistic properties and prevents static segmentation across the corpus, ensuring a fair comparison between deterministic (BPE) and probabilistic (Unigram) subword modeling.

Both subword tokenizers are trained with an identical vocabulary size of 300 tokens on the English source corpus, ensuring that any performance difference between BPE and Unigram reflects the segmentation strategy itself rather than a discrepancy in vocabulary capacity.

6.2 Systems and Baselines

To evaluate transliteration performance, we compare a recurrent neural network with a Transformer-based sequence-to-sequence paradigm. Both models are trained from scratch on our custom vocabularies:

- **BLSTM with Bahdanau Attention:** Selected due to its established success as a state-of-the-art baseline in related Grapheme-to-Phoneme (G2P) tasks. It provides a robust recurrent architecture configured specifically for monotonic phonetic mapping. The encoder uses a single-layer Bidirectional LSTM (hidden dimension of 256) to capture bidirectional grapheme dependencies.
- **BART:** Chosen over models like BERT to fulfill the requirement for a Transformer-based sequence-to-sequence transduction paradigm: in fact, its inherent encoder-decoder structure natively supports the required autoregressive generation. In particular, we used a Transformer encoder-decoder substantially scaled down relative to standard pre-trained BART variants (e.g., BART-base, 140M parameters) to prevent overfitting on our constrained, task-specific dataset. It is configured with a hidden dimension of 256 ($d_{model}=256$), 4 encoder/decoder layers, and 8 attention heads to concurrently attend to multiple orthographic patterns.

Table 3 reports the total parameter count for each architecture alongside the source-side vocabulary

size used by the subword tokenizers. Despite sharing a hidden/model dimension of 256, the two architectures differ substantially in scale: BART’s multi-layer encoder-decoder stack with 8 attention heads yields approximately $7.6\times$ more parameters than the single-layer BLSTM (20.08M vs. 2.65M).

Architecture	Parameters	Src. Vocab
BLSTM + Attn	2.65M	300
BART (Scratch)	20.08M	300

Table 3: Total parameter count and source-side vocabulary size for each architecture (target vocabulary fixed at 25 for both).

Training for both models was carried out in mini-batches of 64 samples, using the AdamW optimizer with a learning rate of 5×10^{-4} for parameter updates.

Preliminary experiments also evaluated GRU (with and without attention) and T5 architectures. These models were explicitly excluded from the final study due to severe training instability and a complete failure to converge, exhibiting oscillating validation error rates that triggered early stopping long before reaching competitive baselines.

6.3 Evaluation Protocol

To ensure a rigorous and reproducible assessment of our transliteration models, we established a strict evaluation framework tailored to the unique challenges of this task.

Custom Test Set and Data Partitioning. Lacking a pre-existing, scientifically rigorous ground truth for this novel cross-lingual task, standard proportionate data splitting (such as an 80-10-10 distribution) was unfeasible. Instead, the synthesized parallel corpus is partitioned into an 80% training split and a 20% validation split using a fixed random seed, while final performance is benchmarked against a manually curated hold-out test set of 300 English nouns. This test set was meticulously transcribed to guarantee a highly reliable evaluation benchmark. To strictly prevent data leakage, approximately 1,200 items, comprising the test words and any training instances sharing their exact morphological prefixes, were systematically purged from the training distribution.

Asymmetric Tokenization Strategy. Our goal was to isolate the effect of source-side processing;

for this reason, subword tokenization (BPE or Unigram) is applied exclusively to the English input. The Italian target sequence remains consistently tokenized at the character level across all experiments. This asymmetric configuration ensures the comparative analysis measures solely how effectively the architecture interprets English orthography, keeping the target generation space invariant.

Training Mechanics and Optimization. Regarding the training mechanics, both architectures are optimized using Cross-Entropy Loss and Teacher Forcing to stabilize gradient descent. The training process is capped at a maximum of 10 epochs, an empirical threshold established to halt optimization just before the training and validation loss curves exhibited the flattening or inversion indicative of overfitting. To further ensure optimal convergence, an early stopping mechanism with a patience of 3 epochs continuously monitors the validation CER.

6.4 Results and Discussion

The empirical performance across the four architectural configurations, evaluated on the independent test set ($N = 300$), is summarized in Table 4. Transliteration fidelity is quantified using the normalized Character Error Rate (CER).

Architecture	Tokenizer	Best Val CER	Test CER
BLSTM + Attn	Unigram	0.0649	0.1113
BLSTM + Attn	BPE	0.0649	0.1224
BART (Scratch)	Unigram	0.0777	0.1262
BART (Scratch)	BPE	0.0772	0.1329

Table 4: Validation and Test Character Error Rate (CER) across model architectures and subword tokenization strategies.

Across all configurations, the Bidirectional LSTM systematically outperformed the Transformer-based BART model, with the **BLSTM + Unigram** combination achieving the lowest overall error rate ($CER = 11.13\%$). This performance hierarchy highlights the importance of inductive bias in phonetic transliteration, which is predominantly a monotonic and local alignment task. The BLSTM naturally enforces sequential recency, making it highly effective on medium-sized datasets ($\sim 107k$ pairs), whereas the standard Transformer decoder, lacking pre-trained priors, showed a higher tendency for phonetic hallucination.

Simultaneously, the Unigram tokenization strategy proved consistently superior to BPE for

both architectures. While BPE’s deterministic merges often fracture irregular phonetic roots into rigid fragments, Unigram’s probabilistic framework preserves phonotactically coherent units. This flexibility allowed the attention mechanisms of both models to learn sharper, more robust alignments with the target Italian character sequences.

6.5 Discussion and Error Analysis

While both paradigms demonstrated high competence in learning the non-linear phonetic mappings between English orthography and transparent Italian graphemes, qualitative examination of the generation logs highlights specific system limitations and recurrent error typologies.

6.5.1 Taxonomy of Committed Errors

The residual errors can be categorized into three primary linguistic phenomena:

1. **Orthographic Interference and Silent Graphemes:** Because English orthography is deeply irregular, words with silent consonants frequently triggered literal character preservation. For example, in words such as *debt* (target: *det*), *doubt* (target: *daut*), and *receipt* (target: *risit*), models occasionally preserved the mute plosive (predicting *debt*, *daubt*, or *risipt*).

2. **Vowel Ambiguity:** A substantial portion of the calculated Character Error Rate arises from unstressed syllables and reduced vowels. Since standard Italian possesses a rigid 7-vowel system without a phonemic schwa, English unstressed vowels can be mapped to multiple acoustically plausible Italian graphemes. For instance:

- *simple* (target: *simpol*) → predicted as *simpal* across models.
- *garden* (target: *garden*) → predicted as *gardan*.

While penalized as character substitutions (*S*) by the CER metric, these variants remain phonetically intelligible and represent valid regional or perceptual approximations.

3. **Non-Local Context and Morphological Outliers:** Extreme orthographic-to-phonetic divergence presented challenges for all models. Notable outliers include toponyms like *Worcestershire* (target:

uusterscier), which resulted in degraded segmentations (e.g., *orsesterscier*, *dorsestersciir*), and loanwords with irregular stress such as *chaos* (target: *cheias* → pred: *ciaus/ciauz*).

6.5.2 Architectural Limitations

Analyzing the behavioral differences between the two model families reveals crucial trade-offs:

- **BLSTM with Bahdanau Attention:** The recurrent architecture demonstrated stronger monotonic alignment and tighter positional bounds, leading to the lowest overall CER. However, its limitation lies in sequential decoding bottlenecks: BLSTM models occasionally suffered from local character duplication or slight truncation when handling long, highly irregular token sequences.
- **BART (Trained from Scratch):** While the Transformer self-attention mechanism excelled at capturing broader contextual dependencies, it exhibited a higher propensity for subtle over-generation and substitution drift (e.g., generating spurious vowel insertions like *chocolate* → *cioucaleit*). Without large-scale self-supervised pre-training, Transformer architectures struggle to develop robust character-level inductive biases solely from small-to-medium task-specific training sets.

Notably, the BART model contains approximately 7.6× more parameters than the BLSTM (20.08M vs 2.65M), yet still underperforms it on this task. This suggests that the BLSTM’s advantage stems from a genuine inductive bias toward monotonic alignment rather than from any capacity constraint. In fact, the massive parameter gap makes the BLSTM’s superior performance an even stronger result. We note, however, that this asymmetry means the comparison is not capacity-matched, and future work could explore a smaller BART variant explicitly matched in parameter count to isolate the architectural effect from the capacity effect.

6.5.3 Dataset and Metric Constraints

Finally, a fundamental limitation stems from the deterministic, rule-based conversion of the CMU Pronouncing Dictionary. Treating natural phonetic variants (e.g., *either*, *garage*) as singular

ground truths inevitably overestimates the error rate of phonetic approximations that a human Italian speaker would consider correct. Furthermore, the test set is notably constrained to approximately 300 instances, a minuscule fraction of the 100,000+ word training corpus. Consequently, the final Character Error Rate (CER) becomes highly sensitive to outliers, potentially misrepresenting the models’ true generalization capabilities. This bottleneck is unavoidable due to the absence of a pre-existing benchmark: creating the reference transliterations necessitated a highly time-consuming, purely manual transcription process by human annotators to ensure linguistic fidelity. Without the possibility of automated generation or reliance on pre-existing resources, scaling the test set to a larger size was prohibitively labor-intensive.

7 Conclusions and Future Work

In this work, we explored the novel task of cross-lingual Grapheme-to-Grapheme transliteration, mapping irregular English orthography into transparent Italian graphemes. Training sequence-to-sequence architectures from scratch using custom subword and character tokenizers established an accessible baseline for this phonetic conversion. Empirically, the Bidirectional LSTM with Bahdanau Attention and Unigram tokenization achieved the lowest Character Error Rate (11.13%). This recurrent architecture effectively captured monotonic phonetic alignments, whereas BART’s self-attention exhibited slight sensitivity to irregular boundaries without large-scale pre-training. Qualitatively, both models mastered core Italian phonotactics, confining residual errors primarily to unvoiced historical consonants and ambiguous schwa reductions.

7.1 Future Work

Building upon the benchmarks and open-source infrastructure established in this study, several promising research avenues emerge:

- **Ground-Truth Expansion and Dataset Balancing:** Curating a more extensive, manually verified test set to improve the distributional balance across the training, validation, and evaluation splits.
- **Integration of Acoustic Information:** Combining lexical representations with acoustic

speech embeddings (e.g., from pre-trained wav2vec models) to resolve orthographically ambiguous homographs (e.g., *lead*, *read*) based on spoken context.

- **Multi-Reference and Perceptual Metrics:** Developing multi-ground-truth evaluation protocols or perceptual loss functions that accommodate valid dialectal and phonetic variants, mitigating metric over-penalization on acceptable vowel approximations.
- **Pre-trained Sequence Models and Low-Resource Transfer:** Investigating the impact of initializing Transformer architectures with multi-lingual pre-trained checkpoints (e.g., mBART, ByT5) to assess whether broad linguistic priors enhance generalization on extreme morphological outliers.
- **Downstream Applications:** Integrating the generated transparent Italian transliterations into interactive second-language (L2) pronunciation learning applications and text-to-speech alignment systems for Italian learners of English.

Contributions. Both authors contributed equally to the conception, execution, and writing of this research, working jointly on the core methodology. Specific primary responsibilities were divided as follows:

- **Gaia Citro:** conducted the state-of-the-art literature review and led the selection of the final comparative models, basing the choices on both existing research and the analysis of initial preliminary tests. Furthermore, she was primarily responsible for manually curating and populating the hold-out test set.
- **Lucia Fornetti:** managed the computational pipeline on Google Colab, overseeing the training and inference phases of the models. Additionally, she refined the deterministic phonetic mapping utilized for the creation of the training dataset.

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